dot(dot(X,Y),Z) → dot(X,dot(Y,Z))

Schematic code
g: graph

A short equation, but replacement dots need new attributes.

Traditional approaches

(a) Algebraic term

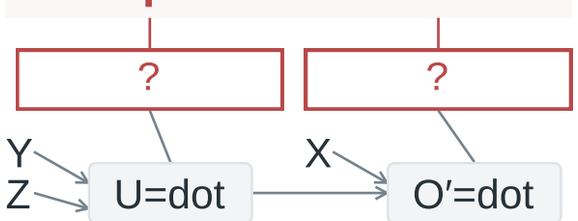
Pattern

```
(dot (dot X Y) Z)
=> (dot X
     (dot Y Z))
```

Supplies the RHS graph, but leaves both dot attributes unspecified.

Attribute handling missing

Incomplete tensor rewrite



Concise rule: ✓

Attributes: ✗

(b) Custom logic

Matching / conditions

```
match_nested_dots(g);
check_conditions(g);
```

Dimension mapping

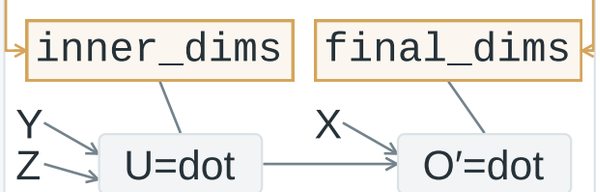
```
inner_dims =
compute_inner_dims(g);
final_dims =
compute_final_dims(g);
```

RHS construction

```
U=build_dot(Y,Z,
  inner_dims);
return build_dot(X,U,
  final_dims);
```

Manual code implementation

Complex and error-prone



Concise rule: ✗

Attributes: ✓ Manual

TEPL

(c) Explicit attributes

Rule

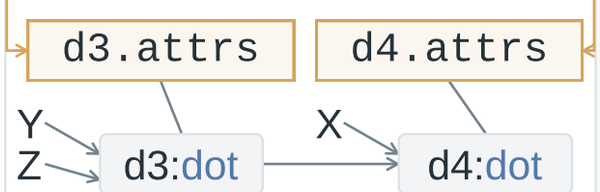
```
(d2: dot
 (d1: dot X Y) Z)
=> (d4: dot X
     (d3: dot Y Z))
```

User-written attributes

```
derive {
  d3.attrs =
    f_U(d1, d2);
  d4.attrs =
    f_final(d1, d2);
}
```

Schematic dimension logic

User-written expressions
Read LHS attributes



Concise rule: ✓

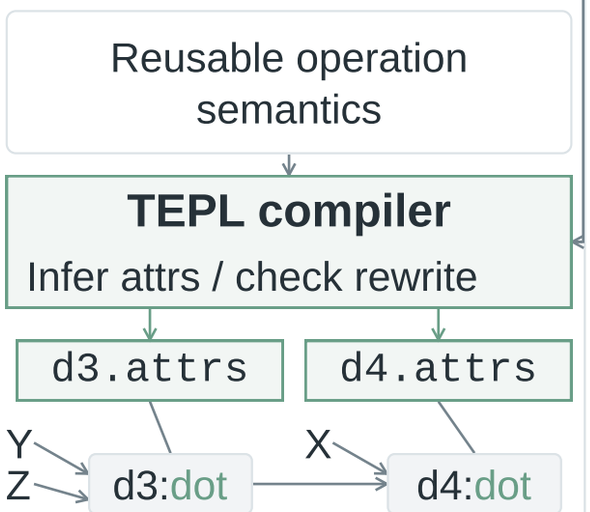
Attributes: ✓ Manual

(d) Inferred attributes

TEPL rule

```
(dot (dot X Y) Z)
=> (dot X
     (dot Y Z))
```

RHS axis fields are inferred from match metadata.



Concise rule: ✓

Attributes: ✓ Automatic